UNIVERSITY OF AGRICULTURE

Department of Computer Science

Lab 6: Programming Fundamentals (CS-308)

Basic Information			
Lab #:	6	**Teacher:**	Arsal Mahmood
Objectives	1. Algorithms, flowcharts, pseudocodes		
Registration#		**Student Name**	
Total Marks	10	**Marks Obtained**	
Tools	Dev C++ / Visual Studio		

Activity 1 : Algorithms, Flowcharts, Pseudocodes

Understand the questions given below and write Pseudocode, Algorithm and draw flowchart of the problems given below followed by code snippet and output.

1. Write a program that calculates the electricity bill according to the following conditions:
   - First 100 units → Rs. 5 per unit
   - Next 200 units → Rs. 7 per unit
   - Remaining units → Rs. 10 per unit
   - The program should input the number of units consumed and display the total bill.

2. Calculate the sum of numbers from 1 to 100 and display it on screen by using for loop.

3. Calculate sum of even and off numbers separately from a user given range by using for loop. Like if user gives lower_limit of 50 and higher_limit of 80. Calculate sum of all odds and evens in between 50 and 80 and display on screen.

4. Write a program that inputs three numbers and determines the largest number using selection statements.

5. Input the marks of a class of ten students. The marks must be between **0** to **100**. Also check marks should not be less than 0 or greater than 100. Calculate the grades of every student and display it by using for loop. Also calculate the total of the grades and the class average.

P-T-O

- >90 = A
- <90 and >80 = B
- <80 and >70 = C
- < 70 and > 60 = D
- <60 = F

6. Calculate sum of even and off numbers by using do-while loop separately from a user given range by using for loop. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and even in between 50 and 80 and display on screen.

7. Print the table of user given number from 1 upto user given limit by using for loop. Like if user enters 3, and limit is 4, program should print table of 3 as follows:

3	*	1	=	03
3	*	2	=	06
3	*	3	=	09
3	*	4	=	12

8. Write a program to find the factorial of a number using a for loop.

9. Write a program that takes an integer and prints its reverse using a for loop.

10. Write a program to check whether a number is a palindrome (same forward and backward) using a for loop.

11. Write a program to calculate x^y (x raised to the power y) using a for loop.

12. Print a diamond shape on screen by using for loop.

```
        *
    *   *   *
*   *   *   *   *
    *   *   *   *
        *   *   *
        *
```

13. Write a program that inputs a number and prints the **reverse of the number** using a loop.

{end of Lab}